

Designing AI Mediators for Eco-Anxiety: Guided Facilitation and Psychological Ownership in Collaborative Journaling

Position paper for IH '26 Workshop: Designing for Negotiation in Collaborative Healthcare

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As the climate crisis exacerbates mental health challenges like eco-anxiety, healthcare systems face unprecedented queues and resource limitations. This position paper explores the role of AI as a *guided facilitator* in collaborative multimedia journaling, designed to extend the reach of clinical systems by promoting guided self-reflection and peer solidarity. Drawing on Value-Sensitive Design (VSD) and Psychological Ownership (PO) theory, we propose an architecture that balances horizontal, cross-border peer-to-peer sharing with stringent institutional adaptation readiness. By framing digital journals as *clinical behaviour settings*, we outline how AI mediators can safely navigate the core value tension between *safe vulnerability* and *data proliferation*. Ultimately, this allows the system to function as a scalable civic resilience infrastructure for multi-stakeholder health ecosystems.

Topics Addressed

- Collaboration, negotiation, and shared decision-making in care
- Participatory, co-design, and user-centered design methods in healthcare research

1 Introduction

Extreme weather events and the broader climate crisis are driving a sharp rise in eco-anxiety, particularly among young adults aged 18–35. This demographic experiences a decline in well-being, disrupted sleep, and heightened anxiety about the future, creating a mental health burden that existing clinical infrastructures lack the speed and reach to effectively cover. For example, in the United Kingdom, mental health issues are a £118 billion economic burden, equivalent to 5% of UK GDP, yet net capital expenditure for mental health fell from £144M in 2015 to £99M in 2022, underscoring a critical gap in support.

Through our recent work on Collaborative Climate Journaling (CCJ), we identified a critical need for scalable, preventative mental health interventions [4]. When clinical pathways are overwhelmed, asynchronous multimedia journaling can offer a therapeutic effect. Furthermore, as we have shown, deliberately structuring these video affordances can transform one-off participation into deeper *relational engagement* [5], empowering individuals to actively articulate and process their emotions. However, to transcend mere symptom-tracking and foster genuine collective resilience, we must investigate how AI mediators can act not as opaque diagnostic filters, but as guided facilitators. In this capacity, AI can actively guide users through structured self-reflection and self-awareness exercises, bridging the gap between clinical capacity and community need, and thereby reducing wait times in mental health queues.

2 Methodological Approach: Uncovering Value Tensions

Designing AI mediators for vulnerable populations requires participatory, user-centred methods to map systemic challenges. Using a Value-Sensitive Design (VSD) approach [3] embedded within a Double Diamond framework [2], our CCJ research engaged mental health professionals and people with lived experiences of extreme weather events.

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This co-design process revealed a core tension inherent in AI-mediated care: *safe vulnerability* versus *data proliferation*. Vulnerable individuals require psychological safety to express raw emotion, yet collaborative care and health system integration demand data liquidity. If an AI mediator simply extracts data to inform clinicians, it risks alienating the user; if it only serves the individual, it fails to integrate into the broader care ecosystem. To resolve this, our proposed architecture utilises *archive-first workflows* and *local, encrypted personal vaults*. Rather than extracting raw video, the AI acts on the device to generate *on-device AI sentiment heatmaps*. This allows the AI mediator to negotiate a shared reality: aggregating individual expressions into population-level clinical data for health professionals and policymakers, without ever exposing the personally identifiable vulnerabilities of the patient.

3 Fostering Relational Engagement via Psychological Ownership

For an AI facilitator to successfully influence health outcomes, users must transition from passive consumption of their health data to active, relational engagement of their lived experiences. We approach this through the lens of Psychological Ownership (PO)—a behavioural science theory that explains why and how individuals feel that something is "theirs". As demonstrated in our recent work on digital advocacy [5], deliberately structuring multimedia interactions around PO mechanisms successfully transforms shallow participation into sustained *stewardship*. Here, we apply this proven lens to evaluate and generate outcomes in relation to patients' health, data, and recovery journeys. PO is a measurable construct that can be used to generate intended health outcomes by evaluating how extrinsic motivations (e.g., automated and personalised journal prompts) can be nudged into intrinsic motivations (e.g., a self-driven desire to maintain emotional hygiene and adhere to a mental self-care routine). To achieve this, the AI mediator must scaffold the three *routes* of Psychological Ownership for the mental health targets:

- (1) **Control:** The AI must allow patients to safely negotiate consent, ensuring they remain in full control over whether, when, and how their journal entries or aggregated insights are shared with the broader community or clinical institutions.
- (2) **Self-Investment:** By prompting users to record multimedia journals, the AI facilitates emotional and time investment, asking for active reflection rather than passive box-ticking.
- (3) **Intimate Knowledge:** The AI acts as a mediator that surfaces relevant peer stories from other patients in a consensually anonymised manner, building community solidarity and a deeper understanding of shared mental burdens like eco-anxiety.

Applying Setting-Driven Design (SDD) [1], we treat the digital journal not as a passive interface, but as a clinical behaviour setting where the AI co-inhabits the space as an active, non-human agent. Rather than acting as a mere tool, the AI functions as a *guided facilitator* that mediates the complex negotiations between a patient's personal vulnerability and the broader clinical ecosystem. To ensure this mediation supports shared decision-making without overriding human autonomy, the AI's interventions must be strictly bounded by *protective frictions* — a critical design requirement to prevent unintended consequences.

4 Framing the Problem Space: Assurance and Institutional Readiness

If an AI mediator is to successfully facilitate shared sense-making or decision-making between citizens, peers, and health systems, we must solve the problem space of *digital trust*. High-stakes video and journaling environments are adversarial; without *assurance-by-design* frameworks, users cannot safely negotiate consent with an AI agent. To address

this, specific *user-held exposure controls* are required that ensure that when patients consent to an AI processing their emotional data, their digital rights and likeness remain fiercely protected.

This assurance infrastructure is what allows the CCJ platform to operate at two distinct scales simultaneously:

- (1) **Horizontal Peer-to-Peer Solidarity:** The system enables cross-border journal sharing, allowing citizens experiencing similar extreme weather event trauma across different geographies to connect, thereby expanding the "circle of we" and reducing isolation.
- (2) **Vertical Institutional Integration:** Simultaneously, the architecture should prioritise being adaptation-ready for institutional and governmental implementation. By structuring the AI's data synthesis to follow strict national regulatory procedures (e.g., GDPR, EHDS), the mediator becomes a safe, compliant tool for regional NHS bodies and public health systems.

5 Conclusion: Toward Transnational Health Ecosystems

The intersection of climate change and mental health requires interventions that transcend traditional clinical boundaries. By designing AI mediators as guided facilitators, we can leverage multimedia journaling to foster self-reflection, reduce clinical bottlenecks, and build measurable Psychological Ownership for patients over their mental health. By balancing borderless peer solidarity with strict adherence to national health regulations, this architecture serves as an exportable standard for large-scale, multi-stakeholder digital health ecosystems. Moving forward, establishing this form of privacy-preserving, AI-mediated care will require transnational, cross-border European collaboration to ensure these civic resilience infrastructures are safe, interoperable, and fundamentally human-centred.

6 Biography and interests

David Sarlos is a design engineer, founder of Vialog Ltd, and will begin doctoral research at Imperial College London's Dyson School of Design Engineering. His work focuses on human-verified, privacy-preserving video ecosystems and behavioural design interventions. Building on his MSc research, the in-development *Journea* project, and Collaborative Climate Journaling work published at ACM Interactive Health 2026, his research explores calibrated reliance in high-stakes human decisions under AI influence. His healthcare interests centre on climate-related mental health, eco-anxiety, and collaborative reflection practices that support patients, peers, and care stakeholders. He is particularly interested in how video journaling and AI can act as a *carefully bounded mediator* in healthcare collaboration: supporting reflection, negotiation, and shared understanding without replacing clinical judgement or human accountability.

References

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